

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$3x(x - 4)(x + 5) = 0$

What is one of the solutions to the given equation?

Answer

- A. -4
- B. 0
- C. 3
- D. 5

Correct Answer: B

Rationale

Choice B is correct. Applying the zero product property to the given equation yields $3x = 0$, $x - 4 = 0$, and $x + 5 = 0$. Dividing each side of the equation $3x = 0$ by 3 yields $x = 0$. Adding 4 to each side of the equation $x - 4 = 0$ yields $x = 4$. Subtracting 5 from each side of the equation $x + 5 = 0$ yields $x = -5$. Therefore, the solutions to the given equation are $0, 4$, and -5 . Thus, one of the solutions to the given equation is 0 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.